

List of publications:

(i) Research Articles:

1. Astha Yadav, Anmol Verma, Saurabh Kumar Singh, **Rohit Prakash***, Bharat Pandey, Sanjay Srivastava, Arun Sethi, Ranvijay Pratap Singh*. " Synthesis of pregn-5-en-3 β -ol-20-one Derivatives via Heck reaction: A Combined Molecular Docking, DFT and NCI-RDG Study". **Journal of Molecular Structure**, **1351 (2026) 144245**.
2. Praveer Singh, **Rohit Prakash***, Saurabh Kumar Singh, Sanjay Srivastava, Arun Sethi, Ranvijay Pratap Singh*. " In Silico Analysis of Methylprednisolone Derivatives: DFT Studies and Molecular Docking Analysis Against Monkeypox Protein". **Letters in Organic Chemistry**, **22 (2025) 465-474**.
3. **Rohit Prakash** Arun Sethi, Sanjay Srivastava, Ranvijay Pratap Singh. " Synthesis, characterization, in silico anti-ovarian cancer activity and quantum chemical calculations of 3- β -(2-(6-methoxynaphthalene-2yl)propionyloxy)-stigmast-5-en". **Indian Journal of Chemistry**, **63 (2024) 1096-1101**.
4. Astha Yadav, Anmol Verma, Saurabh Kumar Singh, **Rohit Prakash**, Sanjay Srivastava, Arun Sethi, Ranvijay Pratap Singh. " Palladium catalysed cross coupling reactions on 2,3-isoxazol-17 α -ethynyltestosterone, their anti-cancer activity, molecular docking studies and ADMET analysis". **Steroids**, **212 (2024) 109515**.
5. Praveer Singh, Ranvijay Pratap Singh, **Rohit Prakash**, Poonam Rawat, Priyanka Yadav, Asmita Singh, Astha Yadav, Sanjay Srivastava, Arun Sethi. "Convenient synthesis of phytosteroid derivatives, their characterization, DFT, QTAIM, NCI-RDG and molecular docking: A combined experimental and theoretical approach". **Russian Journal of Organic Chemistry**, **59 (2023) 1382-1396**.
6. Ranvijay Pratap Singh, **Rohit Prakash***, Astha Yadav, Asmita Yadav, Priyanka Yadav, Sanjay Srivastava, Arun Sethi. "Efficient synthesis of fluorinated derivatives from isolated 24-ethyl cholest-5-en-3 β -ol: Characterization, Quantum chemical calculations by DFT and molecular docking". **Asian Journal of Chemistry**, **34 (11) [2022] 2947-2954**.
7. Arun Sethi, Amandeep, Ranvijay Pratap Singh, Rachana Pathak, **Rohit Prakash**, Sonia Sharma & Rajni Kant. "Expeditionous synthesis of isolated steroids-fluorine prodrugs, their single crystal X-ray crysalligraphy, DFT studies and mathematical modeling". **Indian Journal of Chemistry**. **61(2022) 831-841**.
8. Arun Sethi, Praveer Singh, Priyanka Yadav, **Rohit Prakash**, Ranvijay Pratap Singh. " First synthesis of furostans bearing a 1, 3-dioxane ring at C-26, XRD, Hirshfeld surface analysis, DFT studies". **Journal of Molecular Structure** **1247 (2022) 131364**.
9. Arun Sethi, Praveer Singh, Neera Yadav, **Rohit Prakash**, Ranvijay Pratap Singh, Priyanka Yadav, Monisha Banerjee, "Synthesis of novel steroids using Mizoroki-Heck reaction, their spectroscopic analysis, anticancer activity against cervical cancer and DFT studies" **Journal of molecular structure** **1204 (2020) 127512**.

10. Arun Sethi, Ranvijay Pratap Singh, **Rohit Prakash**, Amandeep, "Facile synthesis of corticosteroids prodrugs from isolated hydrocortisone acetate and their quantum chemical calculations". **Journal of molecular structure** **1130** (2017) **860-866**.
11. Arun Sethi, **Rohit Prakash**, "Novel synthetic ester of Brassicasterol, DFT investigation including NBO, NLO response, reactivity descriptor and its intramolecular interactions analyzed by AIM theory". **Journal of molecular structure** **1083** (2015) **72-81**.
12. Arun Sethi, **Rohit Prakash**, Sangeeta Srivastava, Amandeep, Abha Bishnoi, Ranvijay Pratap Singh, "Isolation of Brassicasterol, its synthetic prodrug-crystal structure, stereochemistry and theoretical studies". **Journal of molecular structure** **1070** (2014) **28-37**.
13. Arun Sethi, **Rohit Prakash**, Amandeep, Dolly Shukla, Akriti Bhatia, Ranvijay Pratap Singh "Identification of phytochemical constituents from biologically active pet. Ether and chloroform extracts of the flowers of *Allamanda violacea*". **Asian journal of plant science and research**, **3(4)** [2013] **95-108**
14. Arun Sethi, Akriti Bhatia, Atul Maurya, Anil Pandey, Gitika Bhatia, Atul Shrivastava, Ranvijay Pratap Singh, **Rohit Prakash** "Proficient synthesis of biologically active pregnane derivatives and its glycoside-Experimental and theoretical approach". **Journal of molecular structure** **1052** (2013) **112-124**
15. Arun Sethi, Akriti Bhatia, Gitika Bhatia, Atul Shrivastava, **Rohit Prakash** "One pot synthesis of biologically active pregnane derivatives, their single crystal structure, spectroscopic characterization, and theoretical calculations". **Journal of molecular structure** **1052** (2013) **76-84**.
16. Arun Sethi, **Rohit Prakash**, Akriti Bhatia, Gitika Bhatia, Ashok K Khanna and Swayam Prakash Srivastava "Hypolipidemic, Hypoglycemic and Antioxidant Activities of Flower Extracts of *Allamanda violacea* A.DC. (Apocynaceae)" **Tropical Journal of Pharmaceutical Research**, **11 (4)**, 2012, **595-604**
17. Arun Sethi, Akriti Bhatia, Dolly Shukla, Abhinav Kumar Ravi Sonker, **Rohit Prakash**, Gitika Bhatia "Synthesis of diosgenin p-nitrobenzoate by Steglich method, its crystal structure and quantum chemical studies". **Journal of molecular structure** **1028** (2012) **88-96**.

(ii) Book Chapter:

Rohit Prakash, "Role of Chemical Industries in the Progress of Modern Forensic Science", **Chemical Industries in Industry 4.0 Era 1 ed.**, Amber Publisher, ISBN: 978-81-978833-1-6.

Workshops/Conferences/Seminars/FDP/Webinars (Organized):

1. **Programme Secretary of National Science day**, Organized by Shri Ramswaroop Memorial University in Collaboration with Vigyan Bharti, **28-Feb, 2021**.

2. **Worked as Coordinator in National Science week (22-28, Feb, 2021)** organized by Institute of Natural Science and Humanities, SRMU.
3. **Organizing secretary of National webinar on "UV and Visible spectroscopy"** on **29th Dec-2020** at SRMU.
4. **Organized an International webinar on "Epoxy and curing agents- A versatile family of synthetics"** as convener on **24th June 2020** at SRMU.
5. **Organized a National Seminar on Innovation for Nation: Innovate for New India, funded by UP CST on 3rd March, 2020** at SRMU.
6. **Organized one week workshop on "Important techniques for the characterization of molecules"** as Co-convener from **25th-29th march, 2019** at Shri Ramswaroop Memorial University, Barabanki.
7. **Organizing secretary in National conference on Recent Approaches and Innovations in Chemical Sciences [RAICS2k18], 2nd-3rd November, 2018** at Shri Ramswaroop Memorial University, Barabanki.
8. **Member of Organizing Committee for 4th Faculty Development Programme** on "New modes of learning and teaching in Higher Education", **10th -15th July, 2017**, Organised by: Internal Quality Assurance Cell, Shri Ramswaroop Memorial University, Barabanki.
9. **Member of Organizing Committee for 3rd Faculty Development Programme** on "Quality Enrichment in Higher Education", **11th -16th July, 2016**, Organised by: Internal Quality Assurance Cell, Shri Ramswaroop Memorial University, Barabanki.

Workshops/Conferences/Seminars/FDP/Webinars (Participated):

1. **Attended the International web-conference on "Recent trends in Chemistry and Advancement (RTCE-2021)**, organized by Department of Chemistry, M.S.J. Government PG College, Bharatpur, Rajasthan on **5-6 March. 2021**.
2. **Participated in Online Faculty Development Programme on "ICT-Enabled tools for effective Teaching-Learning-Examination processes in Higher education"** on **16-20, June, 2020** at SRMU.
3. **Attended the International webinar on "Chemistry from Nature to Mankind to Combat COVID-19[CNMCC-19]**, organized by GSRM Memorial PG College in collaboration with Association of Chemistry teachers and SRMU, from **5-7, June-2020**.
4. **Presented poster in International Carbohydrate Conference (CARBO-XXXIV) "Emerging Frontiers in Carbohydrate Chemistry and Glycobiology"** **5th -7th December, 2019**, Lucknow University, Lucknow and published an abstract.
5. **Participated in National workshop on "Application of technical terminology in Science & technology"**, by Commission for Scientific and Technical Terminology, Ministry of Human Resource and Development at SRMU from **23-24 Oct-2019**.
6. **Participated in two day Faculty Development Program on "Advancement in Learning, Teaching and Research towards Academic Excellence"** on **20-23, July, 2019** organized by Institute of Natural Science and Humanities, SRMU.

7. Participated in **5th Faculty Development Programme on " Challenges of Assuring Quality in Higher Education and Research"**, **2nd-7th July, 2018** Organised by: Internal Quality Assurance Cell, Shri Ramswaroop Memorial University, Barabanki.
8. **Presented poster** in **National Symposium on Innovative Methods in Chemistry Education (IMCE-2015), October 8th- 10th 2015**, Lucknow University, Lucknow and published an **abstract**.
9. **Presented poster** in **21st ISCB International Conference (ISCBC-2015), "currents trends in drug discovery and developments"** **25th-28th February 2015**, CDRI, Lucknow and published an **abstract**.
10. Participated in **International seminar on advances in Bio and nano materials** on **17 November 2013** at department of physics University of Lucknow.
11. **Presented poster** in one day national seminar on **Newer trends in physico-chemical techniques** at department of chemistry, Lucknow University on **August 7 2013** and **won second prize**.
12. Attended and **participated in poster presentation** in **"International Conference on Chemistry: Frontiers and Challenges"**, at the department of Chemistry, AMU and published an abstract in **Souvenir and Abstracts** dated **March 5-6, 2011**.
13. Participated in **poster competition** in Seminar on **"Supramolecular and Organometallic Chemistry"** at department of Chemistry, University of Lucknow on **Mar 25, 2011**.
14. Actively participated in two days seminar on **Applications of 'Analytical Techniques for the characterization of Biomolecules'** Held on **Aug 2-3, 2008**, at University of Lucknow, Lucknow.